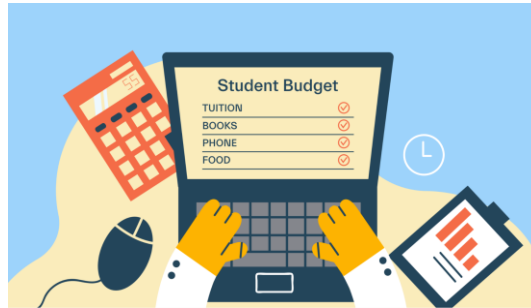


CSC 1300 LAB 8

SPRING 2026

Student Budget



CONCEPTS

- Arrays, 2D arrays
- Functions
- Header files

LAB POLICIES – WHAT YOU CAN'T DO/USE IN THIS LAB

If you do any of the following, you will receive a 0 on some or all of the assignment and possibly get charged with an academic integrity violation (this is very serious).

1. You may not use code or syntax in your lab assignment **that has not been taught** in CSC 1300 class **without referencing your source**. How do you reference your source? By providing a link/URL in a multi-line comment in the section of code where you use “non-1300” code. Otherwise, this will be considered plagiarism. It is safer to use course materials (slides & example programs) to help you create solutions to the assignments.
2. You may not (ever) use the following in your programming & lab assignments in CSC 1300:
 - while(true) or while(1)
 - goto statements
 - recursion
 - exit()
 - ternary operators
 - Any syntax requiring a ‘?’
 - lambda functions
 - try-catch
3. You **may not use generative AI (genAI) in any way** for this lab assignment. Not sure if something is GenAI? Ask a TA or instructor!
4. You may not share your code with anyone other than your lab instructors/TAs/instructors.

LAB REPORT

After you have completed this assignment, fill out the lab report and submit your lab report in your zip file along with your source files. Name your screenshot **lab8ReportProof**.

Lab Report Link: https://tntech.co1.qualtrics.com/jfe/form/SV_6M4wwN1g5Ynv01w

WHAT TO SUBMIT IN ILEARN WHEN YOU ARE DONE

Create a **zip file** named **Lab8_yourTTUUsername** containing **Driver.cpp**, **Functions.cpp**, and **Budget.h** and upload it to the ilearn **LAB 8** assignment folder in your LAB ilearn class.

Replace “yourTTUUsername” with **YOUR** TTU username (the part before @tntech.edu).

SPECIFICATIONS

Write a program that will help **3 college students** keep track of their budget for the month of April.

I am dedicating this lab to those in Sam's lab section who keeps writing gambling programs.

Information Stored for Each Student

- Student's **name** (3-element string array)
- Student's monthly **income** (3-element double array)
- Student's ending **remaining balance after expenditures** for the month (3-element double array)
- **Expenditure** on **four** categories (example of category is food)
 - Note: this should be implemented as a 2-D double array (4 rows, 3 columns)

Name	0	1	2
	Anthony		
Income			
	1000.00		
Ending Balance			
	262.50		
Expenditures	0	1	2
Food	310.00		
Gas	200.50		
Entertainment	510.75		
Phone	120.75		

The categories (like food, gas, entertainment, and phone) will also be entered in by the user and there must be at least 4. I do not recommend having a gambling category.

You can enter this into a 4-element string array or earn bonus points (look below).

Opportunity For Bonus Points

+5 bonus points will be awarded if you store the category names in character arrays (c-strings) instead of strings, which would require a 2-D char array.

Files

Program must be separated into **three files**:

- **Driver.cpp** – contains the main function only
- **Functions.cpp** – contains all the programmer-defined functions
- **Budget.h** – header file that contains all the #includes for the c++ libraries, global constants, and function prototypes

FUNCTIONS

Main Function

- Create the arrays described above, which will hold each students' information. Also create the array to hold the categories.
- Create variables to hold the highest remaining balance, and lowest remaining balance.
- Call a function to get the user's input.
- Call a function to calculate remaining balance for each student.
- Call a function to get the highest remaining balance.
- Call a function to get the lowest remaining balance.
- Print all the information about the student with the highest remaining balance including the students name, income, expenditure amounts (and what the expenditures were for) and their remaining balance.
- Print all the information about the student with the lowest remaining balance including the student's name, income, expenditure amounts (and what the expenditures were for) and their remaining balance.
- Program **may not** contain pointers, structures, vectors, or classes.

Other Functions

- **getUserInput()** – should get the student's name, income, and what they spent on the four expenditures of your choice (like food, etc.)
- **calculateBalance()** – should subtract the student's expenditures from their income and store the result in the remainingBalance array.
- **getHighest()** – should find and return the index of the student who has the highest remaining balance of all students.
- **getLowest()** – should find and return the index of the student who has the lowest remaining balance of all students.

SAMPLE OUTPUT

User input is highlighted in **yellow**.

```
Welcome to the Student Budget Calculator for the Month of April
```

```
Please enter the expenditure category names.
```

```
NAME OF CATEGORY 1 - food
NAME OF CATEGORY 2 - gas
NAME OF CATEGORY 3 - entertainment
NAME OF CATEGORY 4 - phone
```

```
Please enter each student's data.
```

```
STUDENT 1:
```

```
NAME - Andy Crockett
INCOME - $1000
EXPENDITURES
food - $360
gas - $200.50
entertainment - $56.75
```

phone - \$120.25
STUDENT 2:
NAME - Blue Smith
INCOME - \$1500
EXPENDITURES
food - \$543.33
gas - \$0
entertainment - \$150.77
phone - \$143.34
STUDENT 3:
NAME - Spongebob Squarepants
INCOME - \$550
EXPENDITURES
food - \$432.19
gas - \$29.99
entertainment - \$250.03
phone - \$108.34

The student that has the highest remaining balance of \$662.56 is Blue Smith, who started out with an income of \$1500 and expenditures of food \$543.33 gas \$0 entertainment \$150.77 phone \$143.34.

The student that has the lowest remaining balance of \$-270.55 is Spongebob Squarepants, who started out with an income of \$550 and expenditures of food \$432.19 gas \$29.99 entertainment \$250.03 phone \$108.34.